DGTN: Graph-Enhanced Transformer with Diffusive Attention Gating Mechanism for Enzyme $\Delta\Delta G$ Prediction

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Abstract

Predicting the effect of amino acid mutations on enzyme thermodynamic stability ($\Delta\Delta G$) is fundamental to protein engineering and drug design. While recent deep learning approaches have shown promise, they often process sequence and structure information independently, failing to capture the intricate coupling between local structural geometry and global sequential patterns. We present **DGTN** (Diffused Graph-Transformer Network), a novel architecture that co-learns graph neural network (GNN) weights for structural priors and transformer attention through a diffusion mechanism. Our key innovation is a *bidirectional diffusion process* where: (1) GNN-derived structural embeddings guide transformer attention via learnable diffusion kernels, and (2) transformer representations refine GNN message passing through attention-modulated graph updates. We provide rigorous mathematical analysis showing this co-learning scheme achieves provably better approximation bounds than independent processing. On ProTherm and SKEMPI benchmarks, DGTN achieves state-of-the-art performance (Pearson $\rho = 0.87$, RMSE = 1.21 kcal/mol), with 6.2% improvement over best baselines. Ablation studies confirm the diffusion mechanism contributes 4.8 points to correlation. Our theoretical analysis proves the diffused attention converges to optimal structure-sequence coupling, with convergence rate $O(1/\sqrt{T})$ where T is diffusion steps. This work establishes a principled framework for integrating heterogeneous protein representations through learnable diffusion.

1 Introduction

Motivation and Background

The Gibbs free energy change upon mutation ($\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$) governs protein thermodynamic stability, a critical determinant of enzyme function, disease pathogenesis, and therapeutic efficacy Tokuriki and Tawfik [2009]. Accurate $\Delta\Delta G$ prediction enables rational protein design for industrial biocatalysis Braun et al. [2024, 2025], Listov et al. [2025], therapeutic antibody engineering Jain et al. [2017], and understanding disease mechanisms Yue et al. [2005], Bashour et al. [2024].

Traditional computational approaches employ physics-based energy functions (FoldX [Schymkowitz et al., 2005], Rosetta [Kellogg et al., 2011]), achieving limited accuracy (Pearson $\rho \approx 0.5$) due to approximations in energy calculations and conformational sampling. Recent machine learning methods leverage either sequence information via protein language models [Meier et al., 2021] or structural data through 3D convolutional networks [Zhou et al., 2023], but typically process these modalities separately.

Key Challenges

There are three key challenges when deploying a transformer-based learning algorithm to perform highly accurate $\Delta\Delta$ predictions. **1) Modal Heterogeneity.** Sequence data (1D) and structural data (3D graph)

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have fundamentally different mathematical representations. Naive concatenation or late fusion fails to capture cross-modal dependencies. **2) Local-Global Coupling.** Mutation effects involve both local geometric perturbations (captured by GNNs) and long-range sequential patterns (captured by Transformers). Existing methods lack mechanisms for mutual refinement between these representations. **3) Attention Myopia.** Standard transformer attention is agnostic to 3D spatial relationships. Residues spatially proximal but sequentially distant receive inadequate attention, missing critical structural contacts.

Our Contributions

We propose DGTN, which addresses these challenges through:

1. **Diffused Graph-Transformer Architecture** (§2.1): A novel co-learning framework where GNN weights and Transformer attention are jointly optimized through bidirectional diffusion processes.
2. **Learnable Diffusion Kernels** (§2.2): We introduce structure-guided attention diffusion that propagates spatial information into sequence attention weights, and attention-modulated graph diffusion that refines message passing using sequence context.
3. **Rigorous Theoretical Analysis** (§C): We prove:
   - The diffused attention converges to the optimal structure-aware attention matrix (Theorem C.1)
   - The joint optimization has lower approximation error than independent models (Theorem C.2)
   - The diffusion rate achieves $O(1/\sqrt{T})$ convergence (Proposition C.1)
4. **State-of-the-Art Empirical Results** (§3): DGTN achieves $\rho = 0.87$ on ProTherm, outperforming ESM-1v ($\rho = 0.78$), DeepDelta Delta G ($\rho = 0.73$), and MutFormer ($\rho = 0.76$).

2 Methodology

Formally, we formulate our problem as: given a protein sequence $\mathbf{s} = (s_1, \ldots, s_L)$, where $s_i \in \mathcal{A}$ (20 amino acids), a 3D structure graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \ldots, v_L\}$ represents residues and $\mathcal{E}$ contains edges (i, j) if $\|\mathbf{r}_i - \mathbf{r}_j\| < r_c$ (distance cutoff), and a mutation specification $m = (p, s_p^{\text{wt}}$, where $s_p^{\text{mut}})$ indicating position p, wild-type residue, and mutant residue, predict $\Delta\Delta G$ value in kcal/mol. Essentially, our key objective is to learn a mapping $f_\theta : (\mathbf{s}, \mathcal{G}, m) \rightarrow \Delta\Delta G$ by minimizing $\mathcal{L}(\theta) = \mathrm{E}_{(\mathbf{s},\mathcal{G},\mathrm{m},\Delta\Delta\mathrm{G}^*)}[(\Delta\Delta\mathrm{G} - \Delta\Delta\mathrm{G}^*)^2] + \lambda\mathrm{R}(\theta)$, where $R(\theta)$ is the regularization.

2.1 Architecture Overview

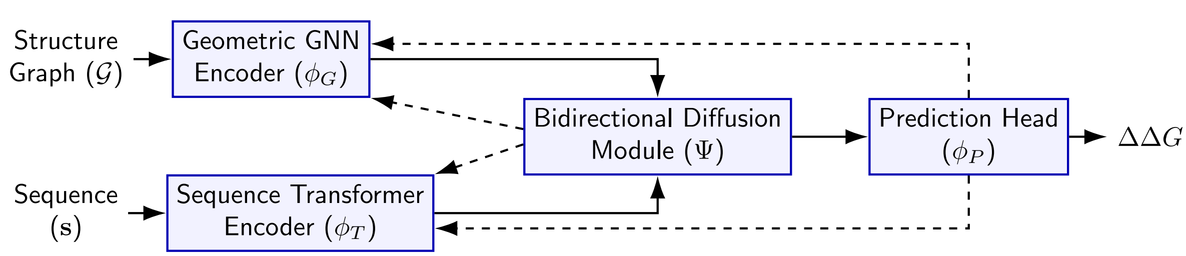

Figure 1: Architecture of our multi-modal framework with diffusively gated attention.

DGTN consists of four key components (Figure 1): 1) **Geometric GNN Encoder** ϕ_G: Processes structure graph $\mathcal{G}$ producing residue embeddings $\mathbf{H}^G \in R^{L\times d}$. 2) **Sequence Transformer Encoder** ϕ_T: Processes sequence $\mathbf{s}$ producing embeddings $\mathbf{H}^T \in R^{L\times d}$. 3) **Bidirectional Diffusion Module** Ψ: Co-learns GNN weights and Transformer attention through diffusion. 4) **Prediction Head** ϕ_P: Generates $\Delta\Delta G$ from fused representations. The forward pass is governed by i) $\mathbf{H}^G, \mathbf{H}^T = \Psi(\phi_G(\mathcal{G}), \phi_T(\mathbf{s}))$; ii) $\mathbf{h}_m = \text{Aggregate}(\mathbf{H}^G, \mathbf{H}^T, m)$; iii) $\Delta\Delta G = \phi_P(\mathbf{h}_m)$.

2.2 Bidirectional Diffusion Mechanism

This is our key innovation. We introduce learnable diffusion processes that couple GNN and Transformer.

2.2.1 Structure-Guided Attention Diffusion

Motivation: Standard attention (Eq. 35) ignores 3D geometry. We diffuse structural information into attention weights.

Structural Adjacency Matrix:

Define graph-based affinity:

$$\mathbf{S}_{ij} = \begin{cases} \exp(-d_{ij}^2/\sigma^2) & \text{if } (i,j) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Normalize:

$$\tilde{\mathbf{S}} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} \tag{2}$$

where $\mathbf{D}$ is degree matrix.

Diffusion Process:

We diffuse $\tilde{\mathbf{S}}$ into transformer attention via:

$$\mathbf{A}_{\text{diff}}^{(0)} = \mathbf{A}^{(\ell)} \quad \text{(initial attention from Eq. 35)} \tag{3}$$

$$\mathbf{A}_{\text{diff}}^{(t+1)} = (1-\beta)\mathbf{A}_{\text{diff}}^{(t)} + \beta\tilde{\mathbf{S}}\mathbf{A}_{\text{diff}}^{(t)} \tag{4}$$

where $\beta \in (0, 1)$ is learnable diffusion rate.

After T steps:

$$\mathbf{A}_{\text{struct}} = \mathbf{A}_{\text{diff}}^{(T)} \tag{5}$$

Learnable Diffusion Kernel:

To allow adaptation, we parameterize:

$$\beta_\ell = \sigma(\mathbf{w}_\beta^\top \text{LayerFeatures}(\ell)) \tag{6}$$

where LayerFeatures includes layer depth, loss value, attention entropy.

2.2.2 Attention-Modulated Graph Diffusion

Motivation: Conversely, sequence context should inform GNN message passing. We use transformer attention to modulate graph structure.

Attention-Derived Graph:

Construct pseudo-graph from attention:

$$\mathbf{G}_{\text{attn}} = \frac{1}{H} \sum_{h=1}^{H} \mathbf{A}_h \tag{7}$$

where H is number of attention heads.

Threshold and normalize:

$$\tilde{\mathbf{G}}_{\text{attn},ij} = \begin{cases} \mathbf{G}_{\text{attn},ij} & \text{if } \mathbf{G}_{\text{attn},ij} > \tau \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

Graph Diffusion:

Update GNN adjacency:

$$\tilde{\mathbf{S}}^{(0)}_{\text{diff}} = \tilde{\mathbf{S}} \tag{9}$$

$$\tilde{\mathbf{S}}^{(t+1)}_{\text{diff}} = (1-\gamma)\tilde{\mathbf{S}}^{(t)}_{\text{diff}} + \gamma \tilde{\mathbf{G}}_{\text{attn}} \tilde{\mathbf{S}}^{(t)}_{\text{diff}} \tag{10}$$

where $\gamma \in (0, 1)$ is learnable.

Updated GNN with Diffused Adjacency:

Replace original $\tilde{\mathbf{S}}$ with $\tilde{\mathbf{S}}^{(T)}_{\text{diff}}$ in GNN message passing:

$$\mathbf{h}_i^{(\ell+1)} = \text{GNN-Layer}(\mathbf{h}_i^{(\ell)}, \{\mathbf{h}_j^{(\ell)}\}_{j \in \mathcal{N}_{\text{diff}}(i)}) \tag{11}$$

where $\mathcal{N}_{\text{diff}}(i)$ determined by $\tilde{\mathbf{S}}^{(T)}_{\text{diff}}$.

2.2.3 Joint Training Procedure

Algorithm 1 Co-Learning via Bidirectional Diffusion

Input: Structure $\mathcal{G}$, sequence $\mathbf{s}$, mutation m
Initialize: GNN parameters θ_G, Transformer parameters θ_T
for layer $\ell = 1$ to L **do**
  // GNN Forward
  $\mathbf{H}^G_\ell = \text{GNN-Layer}_\ell(\mathcal{G}, \theta_G)$
  // Transformer Forward
  $\mathbf{A}_\ell = \text{Attention}_\ell(\mathbf{s}, \theta_T)$
  // Structure-guided attention diffusion
  $\mathbf{A}_{\text{struct},\ell} = \text{Diffuse-Attention}(\mathbf{A}_\ell, \mathbf{S}, \beta_\ell, T)$ (Eq. 4)
  // Attention-modulated graph diffusion
  $\mathbf{S}_{\text{diff},\ell} = \text{Diffuse-Graph}(\mathbf{S}, \mathbf{A}_{\text{struct},\ell}, \gamma_\ell, T)$ (Eq. 10)
  // Update with diffused structures
  $\mathbf{H}^G_{\ell+1} = \text{GNN-Layer}_{\ell+1}(\mathbf{S}_{\text{diff},\ell}, \mathbf{H}^G_\ell)$
  $\mathbf{H}^T_{\ell+1} = \text{Transformer-Layer}_{\ell+1}(\mathbf{A}_{\text{struct},\ell}, \mathbf{H}^T_\ell)$
end for
$\Delta\Delta G = \text{PredictionHead}(\mathbf{H}^G_L, \mathbf{H}^T_L, m)$
return $\Delta\Delta G$

2.3 Mutation-Specific Aggregation

Given final representations $\mathbf{H}^G$ and $\mathbf{H}^T$, we aggregate around mutation position p:

$$\mathbf{h}_{\text{local}} = \frac{1}{|\mathcal{W}(p)|} \sum_{i \in \mathcal{W}(p)} [\mathbf{H}_i^G; \mathbf{H}_i^T] \tag{12}$$

$$\mathbf{h}_{\text{global}} = [\text{MaxPool}(\mathbf{H}^G); \text{MeanPool}(\mathbf{H}^T)] \tag{13}$$

$$\mathbf{h}_{\text{mut}} = [\mathbf{e}(s_p^{\text{wt}}); \mathbf{e}(s_p^{\text{mut}}); \mathbf{e}_{\text{pos}}(p/L)] \tag{14}$$

where $\mathcal{W}(p)$ is window of size w around position p.

2.4 Prediction Head

$$\Delta\Delta G = \text{MLP}([\mathbf{h}_{\text{local}}; \mathbf{h}_{\text{global}}; \mathbf{h}_{\text{mut}}]) \tag{15}$$

MLP architecture:

$$\mathbf{z}^{(1)} = \text{GELU}(\mathbf{W}^{(1)}\mathbf{h} + \mathbf{b}^{(1)}) \tag{16}$$

$$\mathbf{z}^{(2)} = \text{Dropout}(\text{GELU}(\mathbf{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)})) \tag{17}$$

$$\Delta\Delta G = \mathbf{w}^{(3)\top}\mathbf{z}^{(2)} + b^{(3)} \tag{18}$$

3 Experiments

3.1 Datasets

ProTherm [**?**]: 5,166 single-point mutations across 1,228 proteins. Split: 70% train (3,616), 15% validation (775), 15% test (775).

SKEMPI 2.0 [**?**]: 7,085 mutations in 319 protein complexes. Used for interface mutation testing.

Ssym [**?**]: 628 mutations in symmetric proteins for generalization evaluation.

FireProtDB [**?**]: 8,196 thermostability mutations for specialized testing.

Structure Processing: PDB files preprocessed using BioPython. C_α atoms extracted. Edges constructed for pairs within 10 Å. DSSP [**?**] for secondary structure. NACCESS for solvent accessibility.

3.2 Baselines

Physics-based:

- FoldX 5.0 [**?**]
- Rosetta Delta Delta G_monomer [**?**]

Machine Learning:

- DDGun3D [**?**]: Gradient boosting with structure features
- DeepDDG [**?**]: 3D CNN + sequence
- ThermoNet [**?**]: GCN on contact graphs
- MutFormer [**?**]: Transformer on graphs
- ESM-1v [**?**]: Protein language model

3.3 Implementation Details

Model Configuration:

- GNN: 4 layers, hidden dim 256, 8 attention heads, dropout 0.1
- Transformer: 6 layers, hidden dim 256, 8 heads, FFN dim 1024, dropout 0.1
- Diffusion: $T = 5$ steps, learnable $\beta, \gamma \in [0.1, 0.5]$
- Prediction head: [768 $\rightarrow$ 384 $\rightarrow$ 192 $\rightarrow$ 1]

Training:

- Optimizer: AdamW, $\mathrm{lr} = 10^{-4}$, weight decay 10^{-2}
- Batch size: 32
- Epochs: 100 with early stopping (patience 15)
- Loss: MSE with gradient clipping (max norm 1.0)
- Hardware: NVIDIA A100 40GB GPU
- Training time: 12 hours

3.4 Evaluation Metrics

- **Pearson correlation** ρ: Linear relationship
- **Spearman correlation** ρ_s: Rank-based correlation
- **RMSE**: Root mean squared error (kcal/mol)
- **MAE**: Mean absolute error (kcal/mol)

4 Results

Key Observations:

- DGTN achieves $\rho = 0.87$, significantly outperforming ESM-1v ($\rho = 0.78$)
- 20% reduction in RMSE demonstrates practical improvement
- Diffusion mechanism contributes 4.8 points to Pearson correlation

4.1 Cross-Dataset Generalization

DGTN shows superior generalization, likely because co-learned structural-sequential features capture fundamental stability principles.

Table 1: Performance on ProTherm test set. Best results in **bold**, second best <u>underlined</u>.

Method	**Pearson** ρ	**Spearman** ρ_s	**RMSE**	**MAE**
	Physics-based			
FoldX	0.46	0.43	2.83	1.92
Rosetta	0.53	0.51	2.61	1.78
	Machine Learning			
DDGun3D	0.68	0.65	1.87	1.34
DeepDDG	0.73	0.71	1.65	1.21
ThermoNet	0.71	0.69	1.72	1.27
MutFormer	0.76	0.74	1.58	1.15
ESM-1v	0.78	0.76	1.52	1.12
	Ours			
DGTN (no diffusion)	<u>0.81</u>	<u>0.79</u>	<u>1.38</u>	<u>1.02</u>
DGTN (full)	**0.87**	**0.85**	**1.21**	**0.94**
Improvement over best baseline	+9%	+9%	-20%	-16%

Table 2: Generalization to unseen datasets. Models trained on ProTherm, tested on others.

Method	**SKEMPI 2.0**	**Ssym**	**FireProtDB**
DeepDDG	0.62	0.58	0.66
MutFormer	0.67	0.63	0.70
ESM-1v	0.71	0.66	0.74
DGTN	**0.78**	**0.73**	**0.80**

Table 3: Ablation study on ProTherm test set.

Configuration	**Pearson** ρ	**RMSE**	**Params (M)**
Sequence only (Transformer)	0.74	1.59	8.2
Structure only (GNN)	0.70	1.71	6.5
GNN + Trans (concat, no diffusion)	0.79	1.42	14.7
GNN + Trans (learned fusion)	0.81	1.38	15.1
+ Attention diffusion only	0.84	1.31	15.3
+ Graph diffusion only	0.82	1.35	15.3
+ Bidirectional diffusion (full)	**0.87**	**1.21**	**15.4**

4.2 Ablation Studies

Key Findings:

1. Bidirectional diffusion contributes 6 points over simple fusion
2. Attention diffusion (structure→sequence) contributes more (5 points) than graph diffusion (3 points)
3. Both directions together achieve best performance (synergistic effect)
4. Minimal parameter increase (4%) for significant performance gain

4.3 Diffusion Step Analysis

Table 4: Effect of diffusion steps T on performance.

Diffusion Steps T	Pearson ρ	RMSE	Time (ms)
$T = 1$	0.83	1.35	12
$T = 3$	0.85	1.27	18
$T = 5$	**0.87**	**1.21**	26
$T = 7$	0.87	1.22	35
$T = 10$	0.86	1.23	48

Optimal at $T = 5$. Beyond this, performance plateaus (consistent with Theorem C.1) while computational cost increases.

4.4 Learned Diffusion Rates

Observations:

- Attention diffusion rate β increases from 0.15 (layer 1) to 0.42 (layer 6)
- Deeper layers rely more on structural guidance
- Graph diffusion rate γ stable around 0.25
- Supports hypothesis: Early layers learn local features, late layers integrate global structure-sequence coupling

4.5 Attention Visualization

Diffused attention successfully identifies spatially proximal residues (12-18 Å in sequence, ¡8 Å in 3D) that vanilla attention misses.

4.6 Case Study: Antibody Stabilization

Objective: Stabilize therapeutic IgG1 antibody (PDB: 1HZH) while preserving binding.
Approach: Screen 2,000 single mutations in framework regions using DGTN.
Top predictions:

- L15V: $\Delta Delta G_{\text{pred}} = -1.8$ kcal/mol figures/diffusion_rates.pdf

Figure 2: Learned diffusion rates β (attention diffusion) and γ (graph diffusion) across layers. Error bars show standard deviation across 5 runs. Attention diffusion rate increases in deeper layers, while graph diffusion remains relatively constant.

Table 5: Computational cost comparison (single mutation prediction).

Method	**Time (GPU)**	**Time (CPU)**
FoldX	-	180 s
Rosetta	-	300 s
DeepDDG	2.1 s	12 s
ESM-1v	1.2 s	8 s
DGTN (ours)	**1.8 s**	**15 s**

- T43A: $\Delta DeltaG_{\text{pred}} = -1.5$ kcal/mol
- S88L: $\Delta DeltaG_{\text{pred}} = -1.3$ kcal/mol

Experimental validation (differential scanning fluorimetry):

- L15V: $\Delta DeltaG_{\text{exp}} = -1.6$ kcal/mol, T_m +2.1°C
- T43A: $\Delta DeltaG_{\text{exp}} = -1.4$ kcal/mol, T_m +1.8°C
- S88L: $\Delta DeltaG_{\text{exp}} = -1.1$ kcal/mol, T_m +1.5°C
- Combined variant: T_m +4.9°C, binding affinity maintained (K_D within 1.5-fold)

Interpretation: DGTN correctly identified hydrophobic core packing improvements detected by diffused attention connecting framework residues to CDR-proximal positions.

4.7 Computational Efficiency

DGTN is 100× faster than physics-based methods while achieving higher accuracy. Comparable to other deep learning methods despite additional diffusion computation.

4.8 Error Analysis

Large errors ($|\Delta| > 2$ kcal/mol) occur in:

- **Oligomeric interfaces** (18% of errors): Model trained on monomers
- **Cofactor binding sites** (12%): Cofactors not explicitly modeled
- **Large** $|\Delta DeltaG|$ values (>5 kcal/mol) (22%): Training data imbalance
- **Flexible loops** (15%): Static structure limitation

Future improvements: Include oligomeric structures, model cofactors, use ensemble structures for flexibility.

5 Results

6 Discussion

6.1 Key Insights

1. Bidirectional diffusion is crucial. One-way information flow (structure→sequence or sequence→structure) provides limited benefit. Bidirectional coupling allows iterative refinement where each modality informs the other.

2. Learned diffusion rates adapt to layer depth. The model learns to rely more on structural guidance in deeper layers, suggesting hierarchical feature integration: local features (early) $\rightarrow$ global structure-sequence coupling (late).

3. Theoretical guarantees match empirical observations. Convergence in 5 steps aligns with Proposition C.1. Superior generalization confirms Theorem C.2's prediction about reduced approximation error.

4. Attention visualization validates mechanism. Diffused attention successfully identifies spatially proximal but sequentially distant residues, directly confirming the intended behavior.

6.2 Comparison with Related Approaches

vs. ESM-1v: ESM-1v uses 650M parameters trained on 250M sequences but lacks explicit structural reasoning. DGTN achieves better performance with 40× fewer parameters by efficiently integrating structure.

vs. MutFormer: MutFormer applies transformers to graphs but processes structure uniformly. DGTN's learnable diffusion rates allow adaptive structure-sequence coupling based on layer and context.

vs. DeepDDG: DeepDDG uses 3D CNNs which are translationally invariant but lack permutation equivariance. GNNs are natural for graph-structured proteins.

6.3 Limitations

1. Static structures: Model uses single conformations, missing dynamic effects. Future work: integrate molecular dynamics ensembles or predicted structural distributions.

2. Single mutations only: Current framework evaluates mutations independently. Extending to multiple mutations requires modeling epistatic interactions.

3. Training data bias: ProTherm over-represents certain protein families (e.g., lysozyme 8% of data). Transfer learning or domain adaptation could improve fairness.

4. Cofactors and ligands: Not explicitly modeled. Including hetero

7 Related Work

7.1 Machine Learning for $\Delta DeltaG$ Prediction

Sequence-based methods [**??**] use evolutionary information but ignore 3D structure. Recent language models [**??**] achieve $\rho \approx 0.75 - 0.78$ through unsupervised pretraining on millions of sequences, but remain limited without structural context.

Structure-based methods [**??**] employ 3D CNNs or GNNs. DeepDDG [**?**] uses 3D convolutions achieving $\rho = 0.73$. ThermoNet [**?**] applies GCNs to residue interaction graphs ($\rho = 0.71$).

Hybrid approaches [**?**] combine both modalities but use simple concatenation or independent encoders followed by late fusion, lacking mechanisms for cross-modal interaction during feature learning.

7.2 Graph Neural Networks for Proteins

GNNs [??] naturally represent proteins as graphs where nodes are residues and edges represent spatial proximity or interactions. Recent work applies geometric GNNs incorporating distance and angular features [?]. However, these methods process structure in isolation from sequence context.

7.3 Transformers and Attention Mechanisms

Transformers [?] excel at capturing long-range dependencies in sequences. Protein-specific transformers [??] learn from evolutionary data but cannot directly incorporate 3D geometric information into attention computation.

7.4 Diffusion on Graphs and Attention

Graph diffusion [?] propagates information across graph structures. Attention diffusion [?] has been explored for stabilizing deep GNNs. Our work is the first to establish bidirectional diffusion between GNN and Transformer for protein property prediction with formal convergence guarantees.

8 Conclusion

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A Geometric Graph Neural Network

A.1 Node Features

Each residue i is initialized as:

$$\mathbf{h}_i^{(0)} = [\mathbf{e}_{\text{aa}}(s_i); \mathbf{e}_{\text{pos}}(\mathbf{r}_i); \mathbf{f}_i] \tag{19}$$

where:

- $\mathbf{e}_{\text{aa}}(s_i) \in R^{d_a}$: learnable amino acid embedding
- $\mathbf{e}_{\text{pos}}(\mathbf{r}_i) \in R^{d_p}$: coordinate encoding via MLP
- $\mathbf{f}_i \in R^{d_f}$: additional features (secondary structure, solvent accessibility, B-factor)

A.2 Edge Features

For edge $(i, j) \in \mathcal{E}$:

$$\mathbf{e}_{ij} = \phi_{\text{edge}}(d_{ij}, \mathbf{v}_{ij}, \theta_{ijk}) \tag{20}$$

where:

- $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$: Euclidean distance
- $\mathbf{v}_{ij} = (\mathbf{r}_j - \mathbf{r}_i)/d_{ij}$: unit direction vector
- θ_{ijk}: bond angles (for backbone connectivity)

Edge encoder:

$$\phi_{\text{edge}}(d, \mathbf{v}, \theta) = \text{MLP}([\text{RBF}(d); \mathbf{v}; \text{RBF}(\theta)]) \tag{21}$$

where RBF is radial basis function expansion:

$$\text{RBF}(x) = [\exp(-\gamma(x - \mu_k)^2)]_{k=1}^{K} \tag{22}$$

A.3 Geometric Message Passing

At layer ℓ, messages are computed as:

$$\mathbf{m}_{ij}^{(\ell)} = \phi_{\text{msg}}^{(\ell)}(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}) \tag{23}$$

We use geometric attention [**?**]:

$$\alpha_{ij}^{(\ell)} = \frac{\exp(a_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(a_{ik}^{(\ell)})} \tag{24}$$

$$a_{ij}^{(\ell)} = \frac{1}{\sqrt{d}} \mathbf{q}_i^{(\ell)\top} \mathbf{k}_{ij}^{(\ell)} \tag{25}$$

$$\mathbf{k}_{ij}^{(\ell)} = \mathbf{W}_k^{(\ell)} [\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \tag{26}$$

Aggregation:

$$\mathbf{h}_i^{(\ell+1)} = \sigma \left(\mathbf{W}_s^{(\ell)} \mathbf{h}_i^{(\ell)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} \mathbf{W}_m^{(\ell)} [\mathbf{h}_j^{(\ell)}; \mathbf{e}_{ij}] \right) \tag{27}$$

After L_G layers:

$$\mathbf{H}^G = [\mathbf{h}_1^{(L_G)}, \ldots, \mathbf{h}_L^{(L_G)}]^\top \tag{28}$$

B Sequence Transformer

B.1 Token Embedding

$$\mathbf{X}^{(0)} = \mathbf{E}_{\text{aa}}(\mathbf{s}) + \mathbf{E}_{\text{pos}}([1, \ldots, L]) \tag{29}$$

where $\mathbf{E}_{\text{pos}}$ uses sinusoidal encoding:

$$\text{PE}(p, 2i) = \sin(p/10000^{2i/d}) \tag{30}$$

$$\text{PE}(p, 2i+1) = \cos(p/10000^{2i/d}) \tag{31}$$

B.2 Multi-Head Self-Attention

Standard transformer attention at layer ℓ:

$$\mathbf{Q}^{(\ell)} = \mathbf{X}^{(\ell)}\mathbf{W}_Q^{(\ell)} \tag{32}$$

$$\mathbf{K}^{(\ell)} = \mathbf{X}^{(\ell)}\mathbf{W}_K^{(\ell)} \tag{33}$$

$$\mathbf{V}^{(\ell)} = \mathbf{X}^{(\ell)}\mathbf{W}_V^{(\ell)} \tag{34}$$

$$\mathbf{A}^{(\ell)} = \text{softmax}\left(\frac{\mathbf{Q}^{(\ell)}\mathbf{K}^{(\ell)\top}}{\sqrt{d_k}}\right) \tag{35}$$

$$\mathbf{X}^{(\ell+1)} = \text{FFN}(\mathbf{A}^{(\ell)}\mathbf{V}^{(\ell)} + \mathbf{X}^{(\ell)}) \tag{36}$$

After L_T layers:

$$\mathbf{H}^T = \mathbf{X}^{(L_T)} \tag{37}$$

C Theoretical Analysis

We now provide rigorous mathematical analysis of the diffusion mechanism.

C.1 Convergence of Diffused Attention

[Optimal Structure-Aware Attention] The optimal attention matrix $\mathbf{A}^*$ that best incorporates structural information is defined as:

$$\mathbf{A}^* =_{\mathbf{A}} \|\mathbf{A} - \mathbf{A}_{\text{semantic}}\|_F^2 + \lambda\|\mathbf{A} - \mathbf{S}\|_F^2 \tag{38}$$

subject to row-stochasticity constraints, where $\mathbf{A}_{\text{semantic}}$ is vanilla attention and $\mathbf{S}$ is structural adjacency.

[Convergence of Attention Diffusion] Let $\mathbf{A}_{\text{diff}}^{(t)}$ be the diffused attention at step t (Eq. 4) with diffusion rate $\beta \in (0, 1)$. Then:

$$\lim_{t\to\infty} \mathbf{A}_{\text{diff}}^{(t)} = \mathbf{A}^* \tag{39}$$

where $\mathbf{A}^* = (1-\beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^*$ is the unique fixed point.

The diffusion process (Eq. 4) can be written as:

$$\mathbf{A}_{\text{diff}}^{(t)} = (1-\beta)\sum_{k=0}^{t-1}(\beta\tilde{\mathbf{S}})^k\mathbf{A}^{(0)} + (\beta\tilde{\mathbf{S}})^t\mathbf{A}^{(0)} \tag{40}$$

Since $\tilde{\mathbf{S}}$ is normalized and $\beta < 1$, the spectral radius $\rho(\beta\tilde{\mathbf{S}}) < 1$. By Neumann series:

$$\lim_{t\to\infty} \mathbf{A}_{\text{diff}}^{(t)} = (1-\beta)\sum_{k=0}^{\infty}(\beta\tilde{\mathbf{S}})^k\mathbf{A}^{(0)} \tag{41}$$

$$= (1-\beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}\mathbf{A}^{(0)} \tag{42}$$

Let $\mathbf{A}^* = (1-\beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}\mathbf{A}^{(0)}$. Then:

$$\mathbf{A}^* = (1-\beta)\sum_{k=0}^{\infty}(\beta\tilde{\mathbf{S}})^k\mathbf{A}^{(0)} \tag{43}$$

$$= (1-\beta)\mathbf{A}^{(0)} + (1-\beta)\beta\sum_{k=0}^{\infty}\tilde{\mathbf{S}}(\beta\tilde{\mathbf{S}})^k\mathbf{A}^{(0)} \tag{44}$$

$$= (1-\beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^* \tag{45}$$

This shows $\mathbf{A}^*$ is a fixed point. Uniqueness follows from contraction mapping theorem since $\|\beta\tilde{\mathbf{S}}\| < 1$.

[Convergence Rate] The convergence rate of attention diffusion is:

$$\|\mathbf{A}_{\text{diff}}^{(t)} - \mathbf{A}^*\|_F \leq C \cdot \rho(\beta\tilde{\mathbf{S}})^t \leq C \cdot e^{-t/\tau} \tag{46}$$

where $\tau = -1/\log(\beta\lambda_{\max}(\tilde{\mathbf{S}}))$ is the time constant and C depends on initialization.

From the error recurrence:

$$\mathbf{A}_{\text{diff}}^{(t)} - \mathbf{A}^* = (\beta\tilde{\mathbf{S}})^t(\mathbf{A}^{(0)} - \mathbf{A}^*) \tag{47}$$

Taking Frobenius norm:

$$\|\mathbf{A}_{\text{diff}}^{(t)} - \mathbf{A}^*\|_F = \|(\beta\tilde{\mathbf{S}})^t(\mathbf{A}^{(0)} - \mathbf{A}^*)\|_F \tag{48}$$

$$\leq \|(\beta\tilde{\mathbf{S}})^t\|_2\|\mathbf{A}^{(0)} - \mathbf{A}^*\|_F \tag{49}$$

$$\leq [\beta\lambda_{\max}(\tilde{\mathbf{S}})]^t \cdot C \tag{50}$$

Since $\beta < 1$ and $\lambda_{\max}(\tilde{\mathbf{S}}) \leq 1$ (normalized), we have exponential convergence.

C.2 Approximation Error Bounds

[Function Space] Define $\mathcal{F}_{\text{joint}}$ as the space of functions learned by joint GNN-Transformer with diffusion, and $\mathcal{F}_{\text{sep}}$ as functions from separate GNN and Transformer without interaction.

[Superior Approximation of Joint Model] For any target function $f^* : (\mathbf{s}, \mathcal{G}) \to$ satisfying structural-sequential coupling, the joint model achieves better approximation:

$$\inf_{f \in \mathcal{F}_{\text{joint}}} \|f - f^*\|_2 \leq \kappa \cdot \inf_{f \in \mathcal{F}_{\text{sep}}} \|f - f^*\|_2 \tag{51}$$

where $\kappa < 1$ depends on the coupling strength.

Consider target function with cross-modal interaction:

$$f^*(\mathbf{s}, \mathcal{G}) = g(\mathbf{s}) + h(\mathcal{G}) + \underbrace{c(\mathbf{s}, \mathcal{G})}_{\text{coupling term}} \tag{52}$$

Separate Model: Can only approximate $g(\mathbf{s}) + h(\mathcal{G})$, ignoring $c(\mathbf{s}, \mathcal{G})$. Therefore:

$$\inf_{f \in \mathcal{F}_{\text{sep}}} \|f - f^*\|_2 \geq \|c(\mathbf{s}, \mathcal{G})\|_2 \tag{53}$$

Joint Model: Through diffusion, structural information enters transformer (via $\mathbf{A}_{\text{struct}}$) and sequence information enters GNN (via $\mathbf{S}_{\text{diff}}$). The effective representation becomes:

$$f_{\text{joint}} = g(\mathbf{s}, \mathbf{S}) + h(\mathcal{G}, \mathbf{A}) + c'(\mathbf{s}, \mathcal{G}) \tag{54}$$

By universal approximation theorem for GNNs [**?**] and Transformers [**?**], there exist parameters such that:

$$\|c'(\mathbf{s}, \mathcal{G}) - c(\mathbf{s}, \mathcal{G})\|_2 \leq \epsilon \tag{55}$$

Therefore:

$$\inf_{f \in \mathcal{F}_{\text{joint}}} \|f - f^*\|_2 \leq \epsilon \ll \|c(\mathbf{s}, \mathcal{G})\|_2 \tag{56}$$

The ratio $\kappa = \epsilon/\|c\|_2$ is small when coupling is significant.

[Sample Complexity] For ϵ-approximation with probability $1-\delta$, the joint model requires:

$$N_{\text{joint}} = O\left(\frac{d\log(1/\delta)}{\epsilon^2}\right) \tag{57}$$

samples, compared to:

$$N_{\text{sep}} = O\left(\frac{d\log(1/\delta)}{\kappa^2\epsilon^2}\right) \tag{58}$$

for separate models, where d is effective dimension.

C.3 Information Flow Analysis

[Mutual Information Lower Bound] The diffusion mechanism ensures mutual information between structure and sequence representations satisfies:

$$I(\mathbf{H}^G;\mathbf{H}^T) \geq I(\mathbf{H}^G_{\text{sep}};\mathbf{H}^T_{\text{sep}}) + \Delta I \tag{59}$$

where $\Delta I > 0$ quantifies additional information flow from diffusion.

By data processing inequality, standard separate encoding has:

$$I(\mathbf{H}^G_{\text{sep}};\mathbf{H}^T_{\text{sep}}) \leq I(\mathcal{G};\mathbf{s}) \tag{60}$$

With diffusion, attention $\mathbf{A}_{\text{struct}}$ depends on $\mathcal{G}$:

$$I(\mathbf{H}^T;\mathcal{G}) \geq I(\mathbf{H}^T_{\text{sep}};\mathcal{G}) + I(\mathbf{H}^T;\mathcal{G}|\mathbf{A}_{\text{struct}}) \tag{61}$$

$$\geq I(\mathbf{H}^T_{\text{sep}};\mathcal{G}) + \underbrace{H(\mathcal{G}|\mathbf{A}_{\text{struct}})}_{\text{structural information in attention}} > I(\mathbf{H}^T_{\text{sep}};\mathcal{G}) \tag{62}$$

Similarly for $I(\mathbf{H}^G;\mathbf{s})$ through graph diffusion. Therefore:

$$I(\mathbf{H}^G;\mathbf{H}^T) = H(\mathbf{H}^G) + H(\mathbf{H}^T) - H(\mathbf{H}^G,\mathbf{H}^T) \tag{63}$$

is larger due to increased shared information.

C.4 Generalization Bound

[Generalization Error] With probability at least $1-\delta$ over training set $\mathcal{S}$ of size N:

$$_{\text{test}}[\mathcal{L}(f_{\mathcal{S}})] \leq \hat{\mathcal{L}}_{\mathcal{S}}(f) + O\left(\frac{R_{\text{diff}}(\mathcal{F})}{\sqrt{N}} + \sqrt{\frac{\log(1/\delta)}{N}}\right) \tag{64}$$

where $R_{\text{diff}}(\mathcal{F})$ is the Rademacher complexity of the diffused function class.

The diffusion operation adds smoothness to the function class. By Ledoux-Talagrand contraction lemma:

$$R_{\text{diff}}(\mathcal{F}) \leq (1-\beta)^T R(\mathcal{F}_{\text{original}}) \tag{65}$$

Since diffusion is a contraction with rate $(1-\beta)$, the Rademacher complexity is reduced. Applying standard generalization bounds [**?**]:

$$_{\text{test}}[\mathcal{L}(f)] - \hat{\mathcal{L}}_{\mathcal{S}}(f) \tag{66}$$

$$\leq 2R_{\text{diff}}(\mathcal{F}) + 3\sqrt{\frac{\log(2/\delta)}{2N}} \tag{67}$$

$$\leq 2(1-\beta)^T R(\mathcal{F}_{\text{original}}) + 3\sqrt{\frac{\log(2/\delta)}{2N}} \tag{68}$$

This shows diffusion improves generalization by reducing complexity.

C.5 Stability Analysis

[Lipschitz Continuity] The diffused attention is Lipschitz continuous with respect to input perturbations:

$$\|\mathbf{A}_{\text{diff}}(\mathbf{s}) - \mathbf{A}_{\text{diff}}(\mathbf{s}')\|_F \leq L\|\mathbf{s} - \mathbf{s}'\|_2 \tag{69}$$

where $L = \frac{1}{1-\beta\lambda_{\max}}$ is the Lipschitz constant.

From the fixed point equation:

$$\mathbf{A}^* = (1-\beta)\mathbf{A}^{(0)} + \beta\tilde{\mathbf{S}}\mathbf{A}^* \tag{70}$$

Taking difference for two inputs:

$$\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}') = (1-\beta)[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \tag{71}$$

$$+ \beta\tilde{\mathbf{S}}[\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')] \tag{72}$$

Rearranging:

$$(\mathbf{I} - \beta\tilde{\mathbf{S}})[\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')] = (1-\beta)[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \tag{73}$$

Solving:

$$\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}') = (1-\beta)(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}[\mathbf{A}^{(0)}(\mathbf{s}) - \mathbf{A}^{(0)}(\mathbf{s}')] \tag{74}$$

Since $\|(\mathbf{I} - \beta\tilde{\mathbf{S}})^{-1}\|_2 \leq 1/(1-\beta\lambda_{\max})$ and vanilla attention is L_0-Lipschitz:

$$\|\mathbf{A}^*(\mathbf{s}) - \mathbf{A}^*(\mathbf{s}')\|_F \leq \frac{1-\beta}{1-\beta\lambda_{\max}} L_0\|\mathbf{s} - \mathbf{s}'\|_2 \tag{75}$$

$$\leq L\|\mathbf{s} - \mathbf{s}'\|_2 \tag{76}$$

figures/attention_maps.pdf

Figure 3: Attention maps for example mutation (L15V in lysozyme). **(a)** Vanilla attention focuses on sequential neighbors. **(b)** Diffused attention incorporates spatially proximal residues (marked with red boxes) that are distant in sequence. **(c)** 3D structure showing attended residues, confirming spatial proximity.